DATE: 13-10-26

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COURSE NAME: Mobile computing **REG NO**: 192571050

CO1-AT2-Short answer

Q1. Full-duplex voice channels

Formula: Number of full-duplex channels = Total bandwidth / (2 × Simplex channel bandwidth)

Calculation:
N = (45 × 1000) / (2 × 25)
N = 45000 / 50 = 900

Answer: 900 full-duplex voice channels.

Q2. Channels Per Cell

Formula: Channels per cell = Total channels / Cluster size

Calculation:
N = 560 / 7 = 80

Answer: 80 channels per cell.

Q3. GSM Voice Channels

Formula: Total voice channels = Number of carriers × Time slots per carrier

Calculation:
N = 100 × 8 = 800

Answer: 800 simultaneous voice channels.

Q4. Power Ratio in Decibels

Formula: Power ratio (dB) = $10 \log_{10}(P_1 / P_2)$

Calculation:
P = $10 \log_{10}(80 / 4)$
P = $10 \log_{10}(20)$ = 13.01 dB

Answer: 13.01 dB.

Q5. Carrier-to-Interference Ratio

Formula: C/I (dB) = Carrier power − Interference power

Calculation:
C/I = −60 − (−92) = 32 dB

Answer: 32 dB.

Q6. Channels Per Cell

Formula: Channels per cell = Total bandwidth / (Channel bandwidth × Reuse factor)

Calculation:
N = (42 × 1000) / (300 × 7)
N = 42000 / 2100 = 20

Answer: 20 channels per cell.

Q7. Received Power After Doubling Distance

Formula: $P_2 = P_1 \times (d_1 / d_2)^n$

Given: $P_1 = 64$ W, $d_2 = 2d_1$, $n = 4$.

Calculation:
$P_2 = 64 \times (1 / 2)^4$
$P_2 = 64 \times 1/16 = 4$ W

Answer: 4 W.

Q8. Power Required to Double Cell Radius

Formula: Power scaling factor = (New radius / Original radius)²

Calculation:
Factor = $(2R / R)^2$
Factor = $2^2 = 4$

Answer: Four times the original transmit power.

Q9. Total Channels in a Cluster

Formula: Total channels = Number of cells × Channels per cell

Calculation:
N = 4 × 15 = 60

Answer: 60 channels.

Q11. TDMA Slot Duration

Formula: Slot duration = Frame duration / Number of slots

Calculation:
T = 5 / 8 = 0.625 ms

Answer: 0.625 ms per slot.

Q12. Number of Carrier Frequencies

Formula: Number of carriers = Total bandwidth / Carrier spacing

Calculation:
N = (24 × 1000) / 200
N = 120

Answer: 120 carrier frequencies.

Q15. Cluster Call-Handling Capacity

Formula: Cluster capacity = Number of cells × Calls per cell

Calculation:
C = 6 × 250 = 1500

Answer: 1,500 simultaneous calls.

Q17. Total Bandwidth Consumed

Formula: Total bandwidth = Number of channels × Bandwidth per channel

Calculation:
B = 150 × 25 = 3750 kHz
B = 3750 / 1000 = 3.75 MHz

Answer: 3.75 MHz.

Q20. Total User-Handling Capacity

Formula: Total capacity = Regional centers × Base stations per center × Users per base station

Calculation:
C = 8 × 40 × 250
C = 80,000

Answer: 80,000 users.

Q22. Number of FDMA Channels

Formula: Number of channels = Total bandwidth / Channel bandwidth

Calculation:

$N = (6 \times 1000) / 30$

$N = 200$

Answer: 200 channels.

Q23. Offered Traffic in Erlangs

Formula: A = (Number of subscribers × Calls per hour × Average call duration in minutes) / 60

Calculation:

$A = (1200 \times 4 \times 2.5) / 60$

$A = 12000 / 60 = 200$ Erlangs

Answer: 200 Erlangs.

Q24. Maximum Simultaneous Users

Formula: Number of users = Total bandwidth / Bandwidth per user

Calculation:

$N = (40 \times 1000) / 400$

$N = 100$

Answer: 100 simultaneous users.

Q25. Total Network User Capacity

Formula: Total capacity = Number of base stations × Users per base station

Calculation:

$C = 1200 \times 600 = 720,000$

Answer: 720,000 users.

Q26. Paging Rounds Required

Formula: Number of rounds = Total cells / Cells covered per round

Calculation:

N = 200 / 4 = 50

Answer: 50 paging rounds.

Q28. Number of Roaming Subscribers

Formula: Roaming subscribers = (Roaming percentage / 100) × Total subscribers

Calculation:

N = (15 / 100) × 600,000

N = 90,000

Answer: 90,000 subscribers.

Q29. Switching-Center Utilization

Formula: Utilization (%) = (Actual load / Total capacity) × 100

Calculation:

U = (9000 / 12000) × 100

U = 75%

Answer: 75% utilization.

Q30. Channels Per Cell

Formula: Channels per cell = Total bandwidth / (Channel bandwidth × Reuse factor)

Calculation:

N = (48 × 1000) / (200 × 4)

N = 48,000 / 800 = 60

Answer: 60 channels per cell.

Q31. Number of GSM Channels

Formula: Number of channels = Total bandwidth / Channel spacing

Calculation:

N = (30 × 1000) / 200

N = 150

Answer: 150 channels.

Q32. Successful Handoffs

Formula: Successful handoffs = Total attempts × (Success rate / 100)

Calculation:

N = 2500 × (99 / 100)

N = 2475

Answer: 2,475 successful handoffs.

Q34. Cluster User Capacity

Formula: Cluster capacity = Number of cells × Users per cell

Calculation:

C = 7 × 900 = 6300

Answer: 6,300 users.

Q35. TDMA Slot Duration

Formula: Slot duration = Frame duration / Number of slots

Calculation:

T = 4.8 / 8 = 0.6 ms

Answer: 0.6 ms per slot.